



OPEN Optimization of photodegradation of acid blue 1 dye on aluminosilicate supported Cu doped TiO_2 magnetic nanocatalyst using response surface methodology

Fatemeh Mollaie¹, Mahdiah Afroomand², Najmeh Ahmadi³, Nezamaddin Mengelizadeh⁴ & Davoud Balarak⁵✉

In this study, the magnetic TiO_2/Cu /aluminosilicate (MCTA) catalyst was synthesized through a two-step process using the hydrothermal and solvothermal methods. The synthesized materials underwent comprehensive characterization using various analytical techniques. These materials were subsequently applied for the effective elimination of Acid Black 1 dye (AB1). To establish robust correlations among these influential parameters, the research adopted the Box–Behnken design (BBD) within the framework of RSM. This approach ensures a systematic and thorough exploration of the interplay between various variables in the photocatalytic degradation process. In this study, sunlight was utilized as an energy source for the degradation process. The degradation of AB1 reached its peak efficiency under the following optimal conditions: catalyst mass of 0.75 g/L, AB1 concentration of 37.5 mg/L, pH of 3, and a contact time of 45 min. The efficiency, based on the model prediction, was 98.9%, while experimental results showed a 99.1% efficiency. Dark adsorption tests were also conducted, revealing that the dye degradation rate through the photocatalytic process was 18–22 times higher compared to the adsorption process at different concentrations. Toxicity tests revealed that the remaining AB1 in the aquatic environment lost its antibacterial activity after undergoing the studied process. The scavenger experiment provided confirmation that successful photodegradation of AB1 is attributed to the indispensable contribution of both $\cdot\text{OH}$ and $\cdot\text{O}_2^-$. Additionally, MCTA demonstrated outstanding cycle stability in the photocatalytic degradation of AB1. MCTA emerges as a dependable sunlight-responsive heterostructure photocatalyst with significant potential for the photodegradation of pollutants in wastewater treatment.

Keywords RSM, MCTA, Photocatalyst, Adsorption, Acid Black 1 dye, Toxicity tests

Water pollution with toxic and non-degradable chemicals such as heavy metals, artificial dyes, and organic micropollutants threatens the physical and environmental health of billions of people^{1,2}. The estimated usage of dyes is around one million tons per year, widely employed in the production stages of food items, paper, cosmetics, and textiles³. The effluents from these industries contain significant amounts of colorants, leading to long-term environmental pollution, reduced light penetration, decreased dissolved oxygen (DO), increased chemical oxygen demand (COD) in receiving waters, and contamination of groundwater, due to discharge into the environment, even with low concentrations of dyes⁴.

¹Department of Textile Engineering, Islamic Azad University South Tehran Branch, Tehran, Iran. ²Department of Chemistry, Isfahan University of Technology, Isfahan, Iran. ³Student Research Committee, Zahedan University of Medical Sciences, Zahedan, Iran. ⁴Department of Environmental Health Engineering, Faculty of Health, Larestan University of Medical Sciences, Larestan, Iran. ⁵Department of Environmental Health, Health Promotion Research Center, Zahedan University of Medical Sciences, Zahedan, Iran. ✉email: dbalarak2@gmail.com

According to the World Bank research report, textile dyeing and treatment constitute 17–20 percent of industrial water pollution⁵. Approximately, 2000 different chemical substances are used in the textile industry as dyes, and around 15% of these utilized dyes enter the ecosystem along with wastewater⁶. Dyes have diverse effects on human health, including allergies, dermatitis, digestive disorders, and eye irritation⁹. Moreover, dyes are recognized as potential carcinogens that can induce genetic abnormalities. Therefore, the removal of dyes is essential to prevent pollution⁸.

Among different types of dyes, azo compounds are more commonly used due to their color stability and chemical durability. The azo group is classified into reactive, basic, acidic, direct, dispersed, and sulfur dyes based on one or more different azo (N=N) bonds⁹. These dyes are categorized as hazardous substances for the environment due to their toxicity and slow degradation¹⁰. Acidic dyes are typically used in textile industries (nylon, silk, and wool) within the pH range of 3–7. Acidic dyes often contain metal complexes in their structure. AB1, with the formula of $\text{Na}_2\text{C}_{22}\text{H}_{14}\text{N}_6\text{O}_9\text{S}_2$, is an example of an acidic and anionic dye, which is extensively used in textile industries¹¹.

Although pollution resulting from the use of dyes in the textile industry is substantial, it is challenging to reduce it due to the high demand for colored and printed textiles. However, it is possible to decrease the color concentration in the wastewater of these industries before discharge through various processes¹³. There are various physical and chemical methods for removing dyes from aquatic environments. Some of them include adsorption¹⁴, membrane filtration¹⁵, electrocoagulation¹⁶, photocatalytic decomposition¹⁷, and advanced oxidation processes¹⁸. Unfortunately, many of these processes are costly and time-consuming, and they generate a significant amount of sludge, making its disposal challenging. Advanced oxidation processes (AOPs) transform pollutants into harmless substances by generating highly oxidizing free radicals. Consequently, the production of hazardous by-products is avoided¹⁹.

AOPs are classified based on the generation of reactive species, including photocatalysis, sonolysis, ozonation, Fenton oxidation, and electrochemical processes²⁰. Among advanced oxidation processes, photocatalytic oxidation has gained significant attention in the past two decades due to its low operational cost, minimal energy consumption, negligible by-product generation, and environmentally friendly structure, making it an effective technique for the safe detoxification of pollutants present in air and wastewater²¹.

Some materials used as photocatalysts include ZnO ²², CuO ²³, CeO_2 ²⁴, WO_3 ²⁵, and TiO_2 ²⁶. TiO_2 is a semiconductor photocatalyst with strong oxidative power, water affinity, stability, non-toxicity, and insolubility in water. It is utilized for degrading different pollutants, including alkenes PCBs and alkenes PAHs²⁷, alcohol phenols²⁸, and dyes²⁹.

Solar energy technology stands as a form of renewable energy that is both sustainable and environmentally friendly and is abundant and cost-effective. Solar radiation alone is also used for decomposing pollutants in water and wastewater. Nowadays, semiconductor photocatalysts are widely employed in utilizing solar energy to modify toxic dyes, leading to cost reduction in these processes³⁰. The use of TiO_2 as a photocatalyst is constrained by several significant limitations. The accelerated recombination of electron–hole (e^-/h^+) pairs curtails the production of various reactive species such as OH , H_2O_2 , and O_3 during photocatalytic processes³¹. As well, due to the relatively large bandgap of TiO_2 (3–3.2 eV), its photoresponse is limited to the UV range³². Impurities such as metals, non-metals, or common impurities can enhance the photocatalytic performance of TiO_2 ³³. In recent times, there has been a notable advancement in surface modification techniques for nanostructured TiO_2 , involving the incorporation of metals (e.g., Cu, Ag, and Au). This modification has effectively broadened the absorption wavelength of TiO_2 , extending it into the visible region^{34,35}. Among these, copper has been considered as the most effective impurity for the TiO_2 network, significantly reducing the TiO_2 bandgap and shifting absorption from UV to the visible region. Additionally, this metal absorbs electrons generated by light excitation, preventing rapid recombination of e^-/h^+ ³⁶.

The efficacy of TiO_2 as a photocatalyst is contingent upon its surface area, porosity, and crystal structure³⁷. To diminish aggregation and simplify recovery, a diverse spectrum of supports, encompassing zeolites, silica, activated carbon, graphene, and clay minerals, can be effectively employed³⁸. Natural clay minerals possess plentiful adsorption and reaction activation sites, contributing to their outstanding catalytic performance. Their sturdy structure supports effective catalyst recycling, and their unique spatial arrangement allows for the assembly of various dimensional scales. Additionally, their robust cation exchange capacity accelerates catalytic reactions, while surface electronegativity enhances the separation of charge carriers^{39,40}. Aluminosilicate clay minerals (ACM) act as ideal photocatalyst supports, thanks to their expansive specific surface area and abundance of hydroxyl groups on their surface⁴¹.

The separation of these particles from treated water, especially from large volumes of water, is a costly and time-consuming process that has limited its use⁴². Magnetism is a desirable physical property in wastewater treatment because magnetic separation, avoiding centrifugation stages, is an easy method for separating substances from aqueous solutions⁴³. Therefore, the use of TiO_2 on magnetic nanoparticles has been proposed as an efficient solution for the formation of magnetic photocatalysts, offering an effective approach for the separation and useful recovery of photocatalysts⁴⁴.

In view of the minimal studies on the removal of contaminants by TiO_2/Cu /aluminosilicate, our study attempts to progress the photocatalytic performance of TiO_2/Cu /aluminosilicate by implementing a magnetization process. This study also fully investigates all the related processes (mineralization rate and biodegradability, reaction kinetics, toxicity test, catalyst recovery, effects of anions, EPR spectra and scavenging experiments), which is rarely seen in such a complete study.

Therefore, the current investigation endeavors to fabricate a novel nanocomposite through the utilization of the solvothermal method. The synthesized MCTA nanocomposite was employed as a photocatalyst for eliminating AB1. The optimal conditions for maximizing AB1 degradation were determined using the RSM and BBD. The comparative analysis of the photocatalytic performance involved assessing the synthesized

nanocomposite under both sunlight and non-sunlight conditions. Kinetic studies were conducted using the optimized parameters. According to our research findings, we have discovered and reported the degradation of AB1 using the novel MCTA photocatalyst. This significant achievement demonstrates the first documented instance of such degradation using this specific photocatalyst.

Materials and methods

Materials

The chemicals and reagents utilized in this study were sourced from Sigma-Aldrich and adhered to the highest analytical reagent quality standards.

Preparation of Fe_3O_4 nanoparticles

Magnetite cores were first synthesized using the co-precipitation method with the primary substance and Fe^{3+} and Fe^{2+} ions in an alkaline environment. By dissolving 11.66 g of $\text{FeCl}_3 \cdot 6\text{H}_2\text{O}$ and 4.31 g of $\text{FeCl}_2 \cdot 4\text{H}_2\text{O}$ in 200 ml of deionized water, a solution was initially prepared. The solution was placed on a shaker at 100 °C for one hour, and nitrogen gas bubbling was applied to prevent oxidation. In this step, the precipitation of Fe^{3+} and Fe^{2+} occurs in the presence of OH^- . While stirring, 70 ml of 30% ammonia solution was added drop by drop to the reactor. At this stage, the shift in color from orange to brown during this stage signaled the development of iron magnetite oxides (Fe_3O_4) in the solution. At the end of this step, the supernatant was magnetically separated with the assistance of an iron rod.

Preparation of CTA

The CTA nanocomposites were synthesized through the solvothermal method⁴⁵. Initially, in a beaker, bentonite clay (0.5 g) was combined with ethanol (25 ml) and stirred vigorously at room temperature for two hours until a suspension developed. Afterward, 0.2 g of copper chloride was introduced, blending effortlessly with the bentonite and causing a slight color lightening. After an additional hour of stirring, 1.5 g of TiO_2 was added to the solution with ongoing stirring, which turned the mixture greyish-white in color. Following an additional 2 h of stirring, the thick mixture was sealed in an autoclave and heated at 105 °C for 15 min. After centrifugation at 5000 rpm for 15 min, the supernatant was obtained from the resultant mixture. Multiple washes with ethanol were applied to the supernatant, and the product was subsequently dried at 105 °C in an oven for 3 h. The dried product underwent further calcination at 500 °C for 2 h in a furnace to yield the CTA nanocomposite.

Synthesizing MCTA nanocomposites

The synthesis procedure for the nanocomposite is comprehensively depicted in Fig. 1. To produce the final composites (MCTA), a hydrothermal method was employed. Initially, 0.5 g of Fe_3O_4 nanoparticle underwent ultrasonic treatment in 100 mL of deionized water for 2 h. Subsequently, 0.5 g of CTA nanocomposite was added, and the solution underwent further ultrasonication for 1 h. Each solution was then loaded into a 300 ml stainless steel autoclave coated with Teflon and subjected to heating at 200 °C for 12 h, followed by natural cooling to room temperature. The MCTA samples were acquired after washing and drying.

The characterization of the nanocomposite involved various techniques, including UV–VIS Spectrometry (DR5000), FTIR (Thermo Nicolet, USA 370 AVATAR), Scanning Electron Microscopy (TESCAN MIRA2, Czech Republic), XRD (PHILIPS; PW1730), BET (Brunauer–Emmett–Teller), TGA (TA; SDT-Q600), and VSM (MDKB; Iran). Using a potentiostat, EIS was performed with a sinusoidal perturbation voltage set to 2 mV, and the measurements were taken over a frequency range of 0.01 Hz–1 MHz Transient photocurrent.

Photocatalytic and adsorption experiments

The degradation of AB1 solution was employed to gauge the photocatalytic efficiency of MCTA nanoparticles under direct solar radiation, conducted between 10:00 and 14:00, with an average daily temperature of 36 ± 4 °C. Prior to exposure to light, the solutions underwent magnetic stirring in darkness for 60 min to attain a state of adsorption–desorption equilibrium between MCTA and AB1. The preparation of a stock solution involved dissolving AB1 in 1 L of distilled water, resulting in a concentration of 1000 mg/L. Various quantities of the MCTA catalyst were examined for their efficacy in degrading the AB1 solution. Samples were collected at specified time intervals (up to 120 min) and subsequently centrifuged to eliminate the catalyst. The reduction in AB1 concentration was quantified by measuring the absorption at 610 nm using a UV–visible spectrophotometer. The absorbance values were then correlated to concentrations through a standard calibration curve.

Utilizing a TOC analyzer, the levels of Total Organic Carbon (TOC) were assessed. The standard method involving colorimetry and UV–visible spectrophotometry was employed to analyze the initial and final concentrations of Chemical Oxygen Demand (COD). The COD measurement process included adding 2 mL of the wastewater sample to a COD vial containing oxidants, followed by a 2-h digestion at 150 °C. The COD concentration was then determined using a DR-5000 at a specific wavelength. Biochemical Oxygen Demand (BOD_5) levels were measured with a BOD meter (BD 600).

To assess the pH_{PZC} of the catalyst, an appropriate amount of MCTA catalyst was distributed into 11 flasks, each holding 50 mL NaCl solution (0.01 M). The pH range was adjusted between 2 and 12 using NaOH and HCl (0.1 N). Following agitation at 120 rpm for 24 h, the final pH values were documented. The pH_{PZC} was derived by plotting the initial and final pH values and finding their point of intersection.

Experimental design and statistical analysis

In this investigation, the RSM, employing the BBD in three levels (+1, 0, −1), was employed to optimize and analyze the impact of variables, as well as the simultaneous interactions among variables, on the response variable (removal of AB1). Design Expert Version 8 was utilized for this purpose. The variables under examination

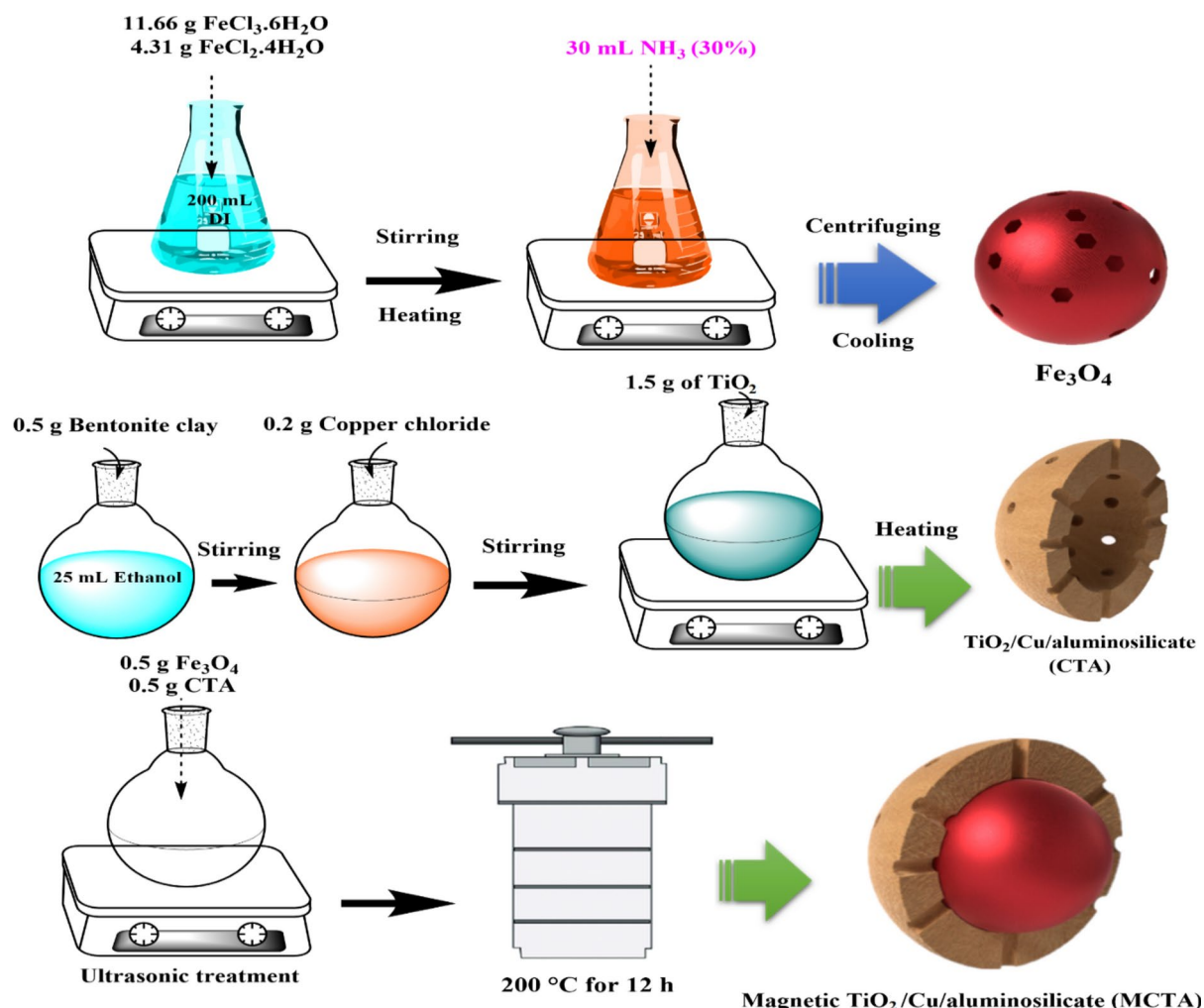


Fig. 1. Illustration of the MCTA synthesis method.

included pH, time, MCTA mass, and the initial AB1 concentration. Utilizing the response level method with the BBD model, the required number of experiments (N) was determined by the equation $N = 2 K(K - 1) + C$; here, K represents the number of variables, while C denotes the number of center points. Considering the removal efficiency of AB1 as the dependent variable (response), Table S1 provides details on the four factors studied, including their chosen levels and associated ranges. To ensure experiment reproducibility, all experiments were conducted with three replications, with a criterion of Relative Standard Deviation (RSD) less than 0.05. Central points were chosen for estimating and gauging experimental error, as well as gauging the lack of fit (LOF). LOF was employed, along with an examination of R^2 and R^2 Adj between laboratory and predicted values, to ascertain the statistical significance of quadratic fit models.

Results and discussion

Characterization of MCTA

The FTIR spectra of the MCTA nanocomposites, as illustrated in Fig. 2a, reveal distinct features. Pre-700 cm⁻¹ absorption peaks signify the bending vibrations of metal oxide bonds, specifically Ti–O at 652.1 cm⁻¹ and Cu–O at 481.4 cm⁻¹, providing confirmation of the successful synthesis of CTA nanocomposites⁴⁶. A pronounced peak at 1053.88 cm⁻¹ specifies the stretching vibrations associated with Si–O–Si bonds. Additionally, the observed band at 3094 cm⁻¹ corresponds to the stretching mode of hydroxyl groups in water molecules adsorbed on the surface⁴⁷. The FTIR spectra also exhibit a characteristic band at 591 cm⁻¹, indicative of the stretching vibration of the Fe–O functional group in magnetite.

The magnetic characteristics of Fe₃O₄ nanoparticles and MCTA nanocomposites were explored, and the findings are depicted in Fig. 2b. The hysteresis loop for Fe₃O₄ nanoparticles demonstrates superparamagnetic behavior, with magnetic saturation occurring around 40 emu/g. Upon the introduction of CTA nanoparticles and the formation of MCTA, there is a decrease in the magnetic saturation of Fe₃O₄ nanoparticles. This reduction can be ascribed to the attendance of CTA nanoparticles within the composite, effectively covering the Fe₃O₄ nanoparticles an observation supported by the hysteresis loop of MCTA as illustrated in Fig. 2b.

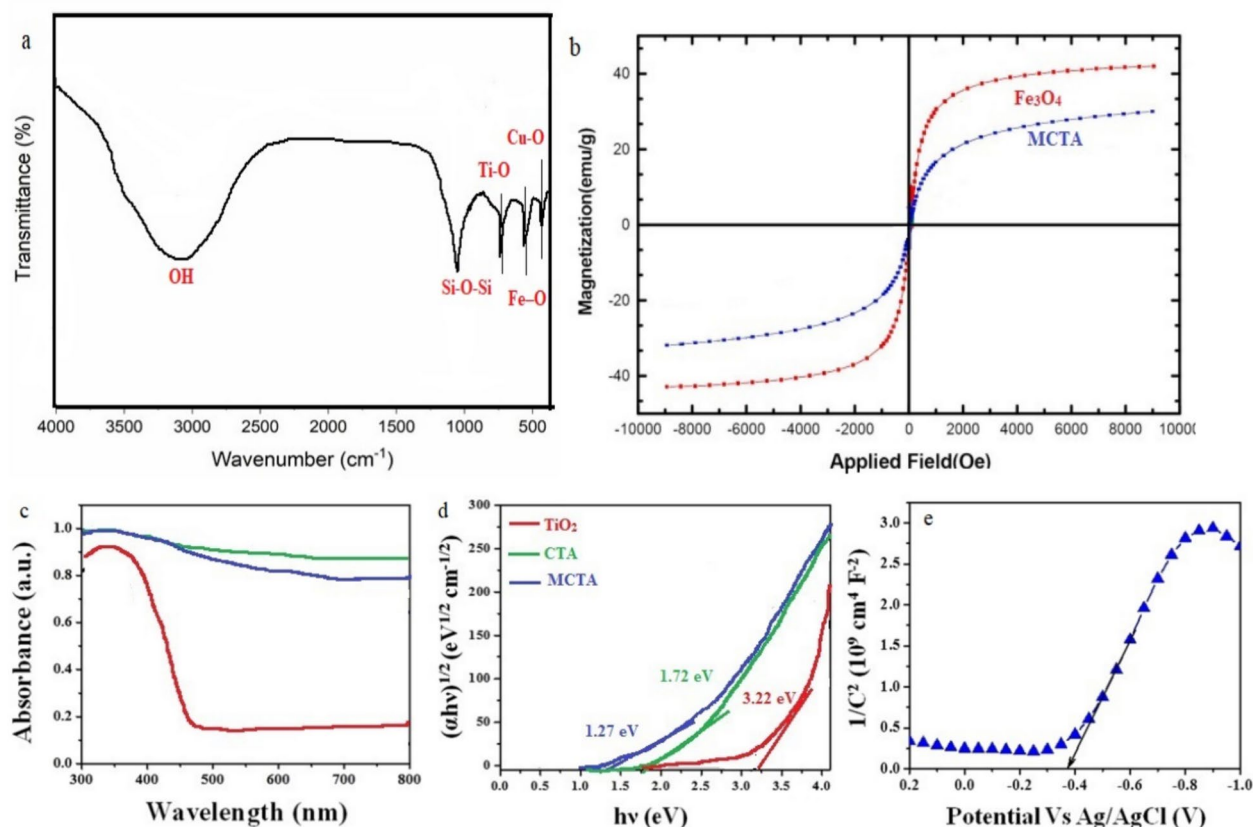


Fig. 2. FTIR analysis (a); VSM curves (b); UV-DRS absorbance spectrum and (c) bandgap spectrum nanocomposites (d); Mott-Schottky plots for MCTA (e).

The diffuse reflectance UV-visible spectra for TiO_2 , CTA, and MCTA are depicted in Fig. 2c. TiO_2 , upon preparation, exhibits heightened absorbance in the UV region, with a peak absorption at around 359 nm. In contrast, a broadened peak ranging from 300 to 800 nm is evident for both CTA and MCTA. This broadening is ascribed to the blue shift of the absorption edges in CTA and MCTA when compared to TiO_2 . Utilizing the Kubelka-Munk function, band gap energy values for the nanocomposites were computed and are presented in Fig. 2d. The plot indicates an intercept line reaching the x-axis, providing the bandgap energy values for the nanocomposites. TiO_2 photocatalyst exhibits a bandgap energy of 3.22 eV, while CTA and MCTA reveal bandgap energies of 1.72 and 1.27 eV, respectively. The reduced bandgap energies in CTA and MCTA, compared to TiO_2 , are attributed to the integration of magnetic nanoparticles and Cu/aluminosilicate materials into the TiO_2 structure. Alterations in size and shape likely contribute to the shift in band gap. The inclusion of magnetic particles facilitates convenient recovery of the nanomaterial from water, and the role of Cu/aluminosilicate is to augment surface area, leading to heightened recombination rates. Ultimately, this enhances pollutant degradation within the specified timeframe.

In Fig. 2e, the Mott-Schottky plots for MCTA are presented. As illustrated, the potentials of the conduction band (CB) in MCTA are recorded at -0.38 eV. Consequently, the valence band (VB) potential is determined to be $+0.89$ eV, calculated based on the CB value and the band gap (1.27 eV) obtained from the optical absorption spectrum.

Figure 3a and b present the data obtained from nitrogen sorption isotherms for bentonite clay and MCTA nanocomposites. Observations in Fig. 3a, corresponding to the adsorption isotherm of bentonite clay, point to the existence of micropores within the material structure. Solids with a microporous structure exhibit a type I adsorption isotherm profile, characterized by rapid nitrogen gas uptake at low relative pressures (p/p_0). An escalation in relative pressure results in the isotherm adopting a type II profile, featuring a notable H2-type hysteresis loop. Bentonite pore volume and BET surface area are determined to be $0.317 \text{ cm}^3/\text{g}$ and $86.9 \text{ m}^2/\text{g}$, respectively. Figure 3b demonstrates that the addition of TiO_2 and Fe_3O_4 to bentonite clay alters the sorption isotherm types of the composites from type I/II to type IV, featuring an $\text{H}_3\text{-H}_4$ hysteresis loop. This change implies the attendance of non-rigid aggregates of plate-like particles, indicating the contribution of micropores and mesopores in MCTA. The BET surface area of the MCTA composite is $95.2 \text{ m}^2/\text{g}$, and the total pore volume is $0.332 \text{ cm}^3/\text{g}$. Additionally, the pore size distribution (PSD) for bentonite clay and nanocomposites was determined using Density Functional Theory (DFT) with medium regularization, as shown in the insert graphics of Fig. 3a and b. These results reveal predominantly mesoporous structures in all solids.

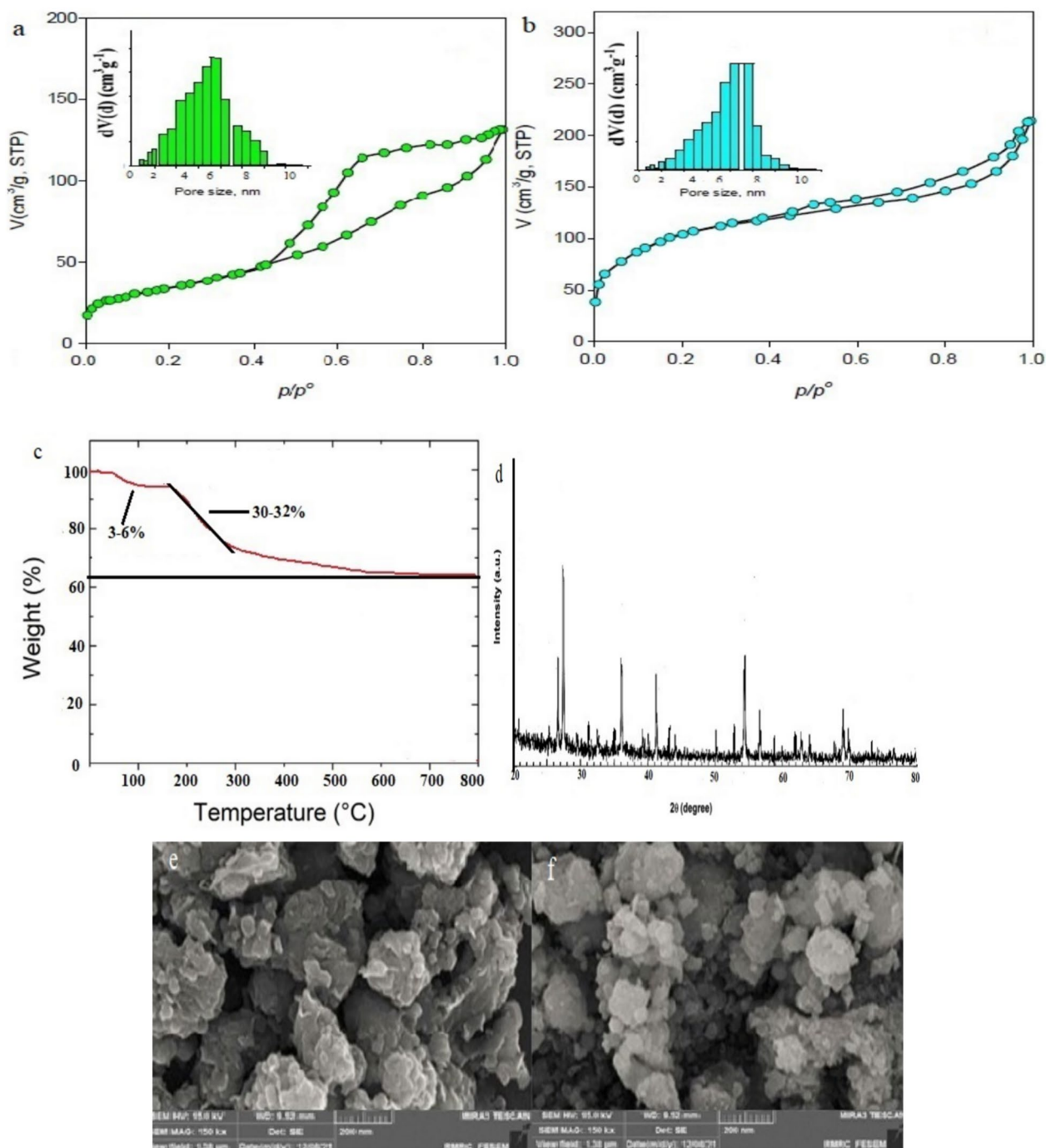


Fig. 3. N₂ adsorption–desorption isotherms and pore size distribution of bentonite clay (a) and MCTA (b); TGA analysis (c); XRD pattern of MCTA nanocomposites (d); SEM images of bentonite clay (e) and MCTA (f).

The thermogram of the MCTA nanocomposites, presented in Fig. 3c, exhibits two distinct stages of thermal weight loss. The initial stage occurs between 60 and 120 °C, resulting in a weight loss of approximately 3–6%. This initial loss is linked to the evaporation of water molecules trapped on the clay's surface. Weight loss in the 60–120 °C typically signifies the removal of solvent molecules, water, or organic matter contained in the sample. At temperatures of 200–250 °C, the second stage of weight loss constitutes about 30–32% of the total weight loss⁴⁵. This stage is linked to the removal of water within the layers of bentonite clay, indicating the dehydroxylation of its layers. Moreover, weight loss beyond 200 °C is associated with decomposition processes.

The XRD analysis for the MCTA nanocomposite is depicted in Fig. 3d. Distinct peaks are evident at angles 26.5°, 27.5°, 36.5°, 41.5°, 54.5°, 56.9°, and 69°, corresponding to the specific diffraction angles of the MCTA nanocomposite⁴⁸. Fe₃O₄ nanoparticles exhibit seven discernible peaks at $2\theta = 31.2^\circ$, 35.58° , 43.5° , 53.7° , 57.5° , 63.2° , and 74.2° , aligning with crystallographic planes (220), (311), (311), (400), (422), (511), (440), and (553),

respectively. Notable peaks at angles 26.5° , 27.5° , 36.5° , and 41.5° indicate the presence of TiO_2 in the sample. Reflection peaks at angles 54.5° and 56.9° suggest the presence of oxygen and silicon atoms (quartz), affirming the existence of bentonite clay. The peak at 69.1° is indicative of copper in the nanocomposite. A sharp peak at 27.5° signifies the anatase phase of TiO_2 , suggesting a reduction in grain size after the synthesis of nanocomposites, consistent with the general trend of decreasing grain sizes of TiO_2 in nanocomposites.

The FESEM analysis images are depicted in Fig. 3e and f. In Fig. 3e, the FESEM image of bentonite reveals particles with a spherical and transparent appearance. Contrastingly, the FESEM image of MCTA in Fig. 3f illustrates the presence of irregularly sized spherical aggregates, appearing to obstruct the bed's surface. These observations are consistent with findings reported in prior studies.

As shown in Fig. 4, photocurrent (PC) and electrochemical impedance spectroscopy (EIS) analyses were conducted to explore the enhanced charge separation ability and the increased lifetime of photogenerated carriers. During repeated 60-s illumination cycles, the electrodes demonstrated prompt and stable current responses. Figure 4a highlights the photoresponse switching patterns of TiO_2 , TiO_2/Cu , and MCTA nanocomposites. The TiO_2 electrode produced a minor photocurrent of $0.22 \mu\text{A cm}^{-2}$, whereas the MCTA nanocomposite generated a photocurrent density of $0.84 \mu\text{A cm}^{-2}$, nearly 3.8 times higher than that of TiO_2 ⁴⁵. The MCTA nanocomposite exhibited a sharp photocurrent increase to $1.35 \mu\text{A}$ under illumination, stabilizing near $1 \mu\text{A}$ before dropping drastically to its original level when the light was turned off. This behavior highlights the substantial improvement in the photocurrent of the quaternary MCTA nanocomposite, with enhanced charge separation efficiency and a prolonged lifespan compared to other materials. To emphasize the benefits of the quaternary MCTA nanocomposite compared to TiO_2 and TiO_2/Cu samples, Nyquist plots were prepared and analyzed, as shown in Fig. 4b. The MCTA nanocomposite exhibited the smallest impedance arc radius, which reflects a lower charge-transfer resistance. This feature is linked to more efficient separation of photogenerated e^-/h^+ and an overall improvement in the photocatalytic performance of the material.

Box–Benken statistical analysis

Selecting an appropriate model for the system, capable of predicting results with high and accurate precision, is a crucial initial step in result analysis. In this context, the proposed quadratic model was employed. The model encompasses four terms representing single effects (A, B, C, and D), six terms for interaction effects (AB, AC, AD, BC, BD, and CD), and four terms associated with quadratic effects (A^2 , B^2 , C^2 , and D^2). To determine the significance of the model and assess its validity, an ANOVA *P*-Value analysis was conducted. The ANOVA results concerning the removal of AB1 are reported in Table 1. The *P*-value for the proposed model, as indicated by the data in this table, is less than 0.0001, suggesting the statistical significance of single effects and their effectiveness (*P*-Value < 0.05). Considering the remaining parameters in the system, Eq. (1) was derived to forecast the AB1 removal efficiency. It encompasses a combination of single, interaction, and quadratic effects.

$$Y = +42.24 - 17.1A + 14.16B + 12.8C - 8.25D - 4.45AB + 2.25AC - 5.25AD + 9.64BC + 1.72BD + 0.825CD - 0.456A^2 + 1.21B^2 + 0.945C^2 + 0.714D^2 \quad (1)$$

In the presented equation:

Y represents the anticipated AB1 removal efficiency in percentage, *A* signifies the pH, *B* corresponds to the quantity of nanocomposite (g/L), *C* denotes the time duration (min), *D* represents the initial concentration of AB1 (mg/L).

The proposed model performance was assessed using key metrics, including correlation coefficient (R^2), adjusted correlation coefficient ($R^2 \text{ Adj}$), and predicted correlation coefficient ($\text{Pred } R^2$). The model demonstrated

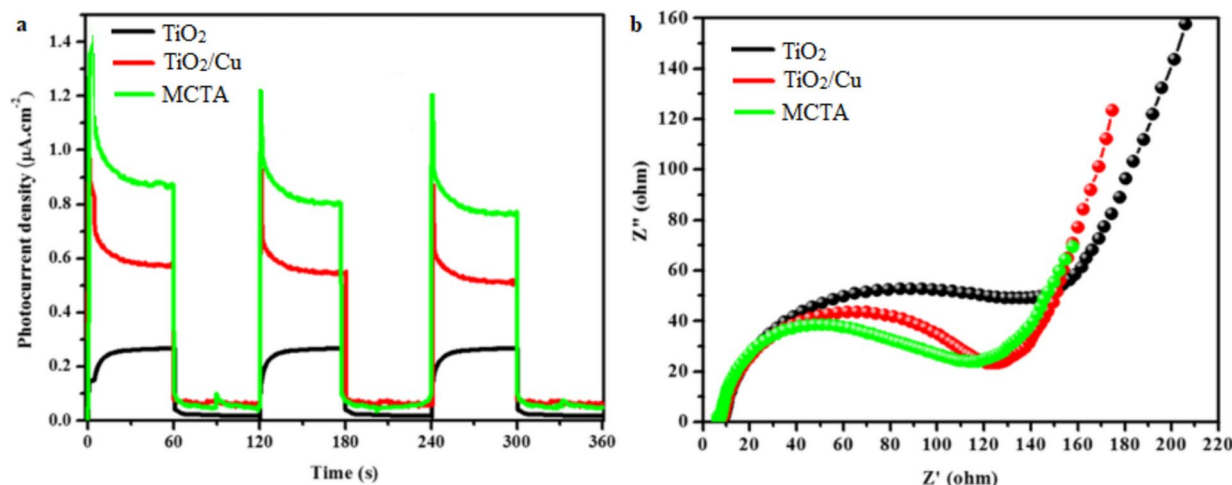


Fig. 4. Transient photocurrent studies (a) and EIS spectra of pristine TiO_2 and the TiO_2/Cu and MCTA catalysts (b).

Source	Sum of Squares	df	Mean Square	F Value	p-value (Prob > F)	
Model	19,524.2	15	1156.2	85.2	<0.0001	Significant
A-pH	11.2	1	11.2	17.2	<0.0001	
B-Dose (g/L)	18,421.2	1	18,421.2	854.2	<0.0001	
Contact time (min)	5563	1	5563	126.2	<0.0001	
D- Concentration (mg/L)	109.2	1	109.2	35.1	0.0001	
AB	5.65	1	5.65	3.44	0.0045	
AC	2.25	1	2.25	0.0019	0.981	
AD	3.15	1	3.15	0.232	0.841	
BC	11.2	1	11.2	2.89	0.004	
BD	0.956	1	0.956	0.246	0.741	
CD	0.825	1	0.825	0.0048	0.902	
A ²	0.641	1	0.641	3.74	0.005	
B ²	4825.3	1	4825.3	442.3	<0.0001	
C ²	38.2	1	38.2	24.1	0.0084	
D ²	0.0714	1	0.0714	9.25	0.0894	
Residual	135.2	14	115.3			
Lack of Fit	110.2	6	16.2	3.94	0.0741	Not significant
Pure Error	11.2	4	5.65			
Cor Total	19,852.1	28				

Table 1. Results of ANOVA For AB1 degradation by MCTA.

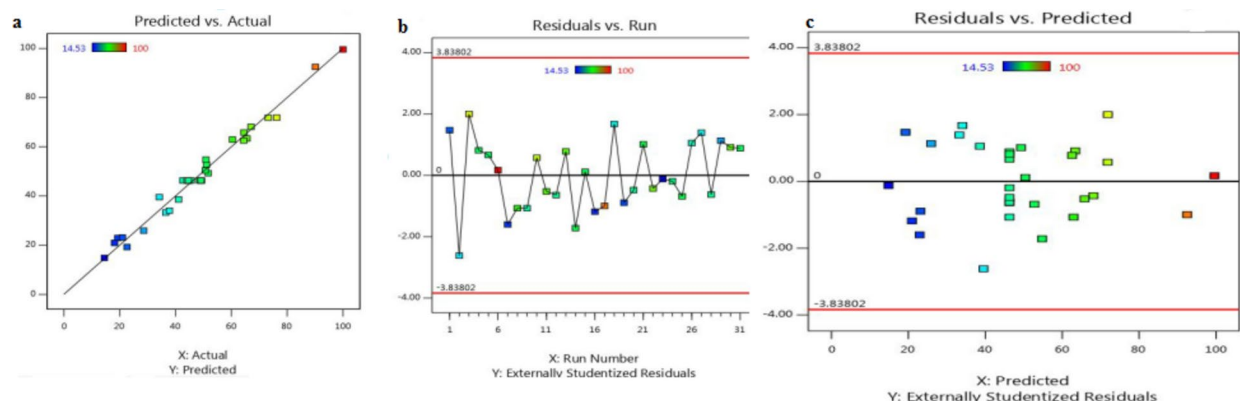


Fig. 5. Illustrating the distribution of the experimental data versus the model's predicted values.

high accuracy with R^2 and R^2 Adj values of 0.992 and 0.981, respectively. The close alignment of these values indicates the model's robust accuracy. This underscores a robust relationship between the experimental findings and the values predicted by the statistical model, confirming the model's reliability and effectiveness.

In Fig. 5a, the comparison between predicted values and observed values is depicted, while Fig. 5b displays the standardized residues for each run. Analysis of these diagrams suggests that, across all conducted experiments, the standardized residue values fall within an acceptable range. Additionally, Fig. 5c portrays internally studentized residual values, affirming the model acceptance and providing opportunities for variance analysis.

Figure 6 provides a visual representation of the crucial variables influencing AB1 removal, e.g., nanocomposite dose, reaction time, the initial concentration of AB1, and pH. The figure offers insights into the variable ranges, selected intervals, impact magnitudes, and optimal points for each parameter. Al-Musawi et al.⁴⁸, utilizing titanium dioxide nanoparticles/graphene oxide, investigated the sonocatalytic process for eliminating acid orange 7 dye from aqueous solutions. Through the application of RSM, the quadratic model demonstrated superior fitting when analyzing COD removal data, with the solution pH identified as a highly influential parameter ($p < 0.0001$).

The adjusted R^2 solidifies the high significance of the model ($R^2 > 0.98$). The experiments demonstrate high precision and reliability, as evidenced by the low coefficient of variation (C.V) at 2.4. Furthermore, with a p -value < 0.0001 and an F-value of 85.2, the model demonstrates a meaningful correlation. As observed in Table 1, the Lack of Fit for the ultimate models indicates statistical significance, affirming that the models are in strong agreement with the experimental data.

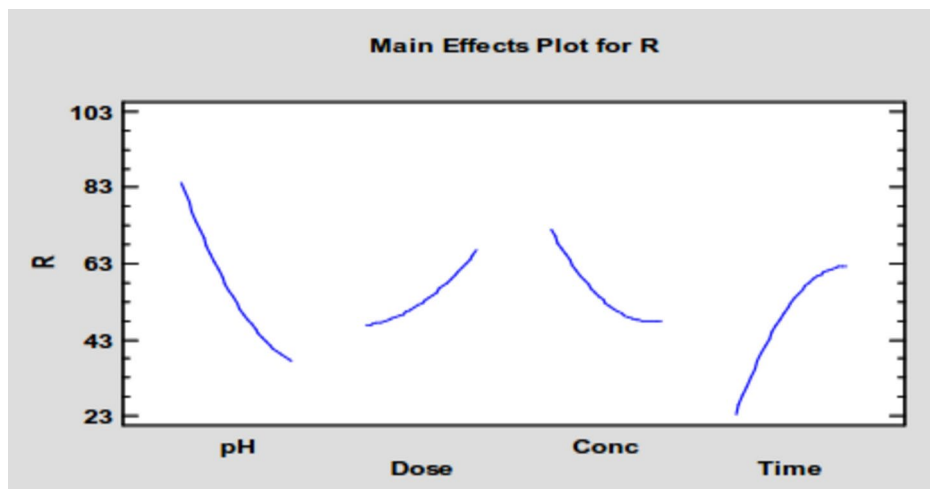


Fig. 6. The role of the initial parameters (pH, dose, concentration, and time) and their optimal points.

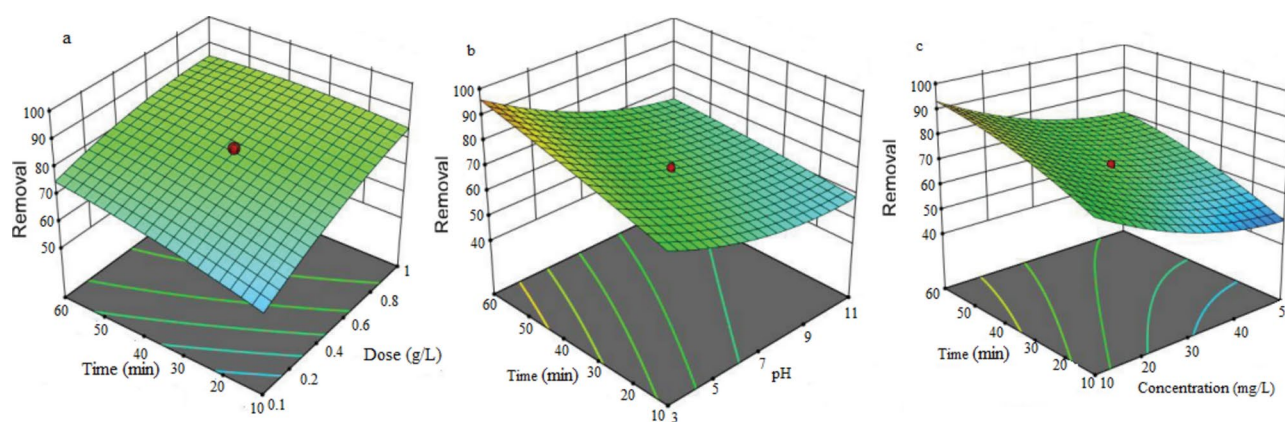


Fig. 7. AB1 degradation by MCTA in a three-dimensional surface plot, illustrating the interactive effects of the four factors under test.

Moreover, the attained Adequate Precision ratio of 24.2, exceeding 4, indicates a strong signal for the second-order model, suggesting its suitability for application within the design space.

Based on the p -value being less than 0.1, factor effects and interactions for AB1 removal by the process in Table 1 indicate the statistical significance of all main factors. The greatest impact on AB1 removal is observed for the catalyst dosage ($F = 854.2$), while the least impact is attributed to the initial pH ($F = 17.2$) among the main factors. Results from Table 1 demonstrate that the overall effects of interactions are less significant than those of the main factors. Of these, Statistical significance is observed for the AB and BC interactions in the model. Regarding second-order interactions, all are statistically significant, with the B^2 interaction having the most substantial impact, due to its high F -value, on the AB1 degradation by the MCTA catalyst.

The optimum operating conditions for the AB1 degradation process are as follows: an AB1 concentration of 37.5 mg/L, a catalyst mass of 0.75 g/L, a time of 45 min, and a pH of 3. The efficacy, based on the model prediction, is 98.9%, while experimental results demonstrate an efficiency of 99.1%.

3D response surface plot analysis

Figure 7a shows the surface plots of AB1 photodegradation. The plot represents the relationship between the catalyst dosage and the corresponding irradiation time. The elliptical pattern of the 3D plot indicates that AB1 photodegradation was influenced by the interaction between the catalyst concentration and the duration of irradiation, as illustrated in the figure. Additionally, an increase in MCTA dosage enhanced the removal efficiency. An elevated catalyst concentration improves the access to active surface sites, allowing for the absorption of additional photons. This results in the formation of higher quantities of $O_2^{\cdot-}$ and $\cdot OH$, facilitating the degradation of more AB1 molecules³⁴. Evidence from studies suggests that rising the catalyst dosage develops the degradation of methylene blue dye. This enhancement is explained by the rise in active photocatalytic sites, which absorb more photons and produce greater quantities of $O_2^{\cdot-}$ and $\cdot OH$ radicals, leading to more effective pollutant removal²⁹. It was apparent that AB1 degradation was maximized after 60 min of reaction. The results

validated the positive influence of extended contact time on the degradation process. Therefore, the enhanced removal of AB1 was attributed to increased contact time, which allowed for a higher probability of interactions between e^-/h^+ pairs and AB1 molecules. Hitkari et al.²³ found that the degradation of congo red dye was significantly enhanced with increasing contact time, rising from 0% to 89.2% between the start and 60 min of the reaction. It was shown by other researchers that naproxen removal efficiency increased with prolonged contact time, as longer contact time generated a greater abundance of active sites on the catalyst surface³⁶.

pH adjustments in the solution are instrumental in governing the dye adsorption process through modifications in the adsorbent's surface charge, pollutant ionization, and dye structure. By raising the contact time and reduction the pH value, as presented in Fig. 7b, the photodegradation efficiency of AB1 increases.

The effect of pH on AB1 adsorption can be described based on the catalyst pH_{zpc} , which is 7 (Fig S1). At pH_{zpc} , the acidic and basic functional groups do not alter the solution pH. For pH below pH_{zpc} , the MCTA surface becomes positively charged, attracting anionic dyes from the solution. When the pH is above pH_{zpc} , the MCTA surface becomes negatively charged, attracting cationic dyes⁵. Therefore, at $pH < 7$, a substantial positive charge density is evident on the MCTA surface, enhancing the adsorption of the AB1 dye, which carries a negative charge. With a high negative charge density on its surface at pH levels above 7, MCTA exhibits decreased adsorption of the negatively charged AB1 dye. The initial pH's effect on dye adsorption is likely attributable to chemical and electrostatic interactions between the adsorbent and the dye molecules¹¹.

Variations in dye concentration, particularly over a contact period of 10–60 min (see Fig. 7c), are linked to the observed decline in photodegradation performance. While an initial increase in AB1 concentration improves the reaction rate by supplying more reactants to the catalyst's surface, further increases in dye concentration reduce photocatalytic efficiency. This reduction results from the higher number of pollutant molecules in concentrated solutions, which block light penetration to the photocatalyst surface. As a result, fewer hydroxyl radicals are generated, limiting their ability to degrade contaminants effectively²⁰. Nevertheless, over time, the adsorption of dye molecules on the catalyst surface hinders the adsorption of oxygen and water molecules, thereby limiting the formation of $\cdot OH$ ¹⁹. Moreover, an increase in AB1 concentration leads to light scattering, while the dye's electrons can become excited upon absorbing energy from the light. Hence, the absorption of emitted light by excited dye molecules increases with rising dye concentration, which consequently lowers the generation of hydroxyl radicals by the photocatalyst²¹.

Reaction kinetics

Pseudo First Order (PFO) or the Hinshelwood-Langmuir model can be employed to aptly characterize the photocatalytic reaction kinetics involving organic compounds. The mentioned model intricately relates the degradation rate to the pollutant concentration, as succinctly expressed in Eq. (2)⁴⁹:

$$r = \frac{dC}{dt} = \frac{k_d KC}{1 + KC} \quad (2)$$

In the given equation, K represents the main constant rate (1/min), and k_d signifies the absorption equilibrium constant (l/mg).

In scenarios where adsorption is relatively low or the concentration of the organic compound is low, Eq. (2) can be reduced to PFO. This form includes a constant rate ($K_{app} = k_d K$, 1/min), as defined in Eq. (3):

$$\ln \frac{C_0}{C_e} = K_{app} t \quad (3)$$

In the mentioned equation, C_0 and C_e refer to the concentrations (mg/L) of AB1 at the beginning and after a designated time (t , min) of the photocatalytic reaction, respectively.

The half-life constant is a pivotal parameter in the pollutant removal process, signifying the time required for the pollutant to degrade by 50%. This constant is determined using the equation represented below⁵⁰:

$$t_{1/2} = \frac{\ln 2}{K_{app}} \quad (4)$$

The results obtained for both processes conform to the first-order kinetics model, exhibiting a high regression coefficient. Table 2 presents the reaction rate coefficients for various concentrations in both the adsorption and photocatalytic processes. Notably, a rise in concentration causes a reduction in the reaction rate coefficient, which matches the observed target percentage. Moreover, the photocatalytic process demonstrates a markedly higher reaction rate coefficient compared to the adsorption process, ranging from 18 to 22 times higher across different concentrations. Based on these findings, the half-life constant is calculated for concentrations from 10 to 100 mg/L for both processes. In the adsorption process, the half-life constant exhibits an increasing trend with concentration, ranging from 96.2 to 288.7 min. In contrast, the corresponding half-life constant for the photocatalytic process is considerably shorter, varying from 5.13 to 13.32 min for the same concentrations. The adsorption rates for concentrations of 10, 25, 50, 75, and 100 mg/L are 4.6, 9.25, 13.2, 20.4, and 24.6 mg/g, respectively. Comparable findings were seen in research by Al-Hawary et al.⁵¹, where the photocatalytic process exhibited a reaction rate coefficient tenfold higher compared to the adsorption process during the degradation of acid red 14, with respective half-life constants of 11.2 and 121.4 min. Additionally, Yilmaz et al.²⁰ observed comparable trends in their study on the degradation of acid blue 113 by $Fe_3O_4/ZnO/GO$, reporting a greater reaction rate coefficient for the photocatalytic process compared to adsorption.

Adsorption processes				Photocatalytic processes			
C ₀ (mg/L)	K _{app} (min ⁻¹)	R ²	t ^{1/2} (min)	C ₀ (mg/L)	K _{app} (min ⁻¹)	R ²	t ^{1/2} (min)
10	0.0072	0.981	96.2	10	0.135	0.981	5.13
25	0.0065	0.912	106.6	25	0.119	0.972	5.82
50	0.0051	0.934	135.9	50	0.094	0.992	7.37
75	0.0032	0.952	216.5	75	0.071	0.954	9.76
100	0.0024	0.931	288.7	100	0.052	0.938	13.32

Table 2. The kinetic properties of the PFO equation with regard to the breakdown of AB1.

Effects of anions

Anionic species hold considerable significance in the chemical composition of aqueous systems (63). The photodegradation of AB1 using the MCTA photocatalyst is analyzed in this section, focusing on the impact of anions. Specifically, 0.1 M solutions of Na₂CO₃, NaCl, NaNO₃, Na₃PO₄, and Na₂SO₄ were individually employed to assess how carbonate, chloride, nitrate, phosphate, and sulfate ions impact the reaction rate. The experiments utilized an initial concentration of 37.5 mg/L of AB1 with 0.75 g/L of MCTA. As depicted in Fig. 8a, the presence of these anions marginally inhibited the degradation of AB1, resulting in prolonged reaction times. This inhibition effect aligns with previous observations involving anionic species, e.g., NaCl, NaNO₃, and Na₂SO₄⁵². Competing for active sites on the photocatalyst surface or inducing deactivation is a potential consequence of the presence of these anions. Additionally, some studies propose that these anions may function as scavengers for active species (h⁺ and •OH), generating less reactive radicals such as Cl• or ClOH•• (Eqs. 5 and 6). The formation of these fewer active radicals subsequently decelerates the catalytic degradation rate^{53,54}.



Mineralization rate and biodegradability

The effectiveness of an oxidation process in eliminating organic matter is significantly influenced by its mineralization and biodegradability. In this study, the mineralization rate of AB1 in the MCTA catalyst was assessed using COD and TOC indices. As depicted in Fig. 8b, the TOC removal efficiency escalates from 0 (41.2 mg/L) to 91.1% (3.67 mg/L) with an increasing reaction time of 0 to 120 min, while the AB1 removal rate remains steady at 99.1%. These findings corroborate the COD removal results, indicating a reduction in COD from 117.4 to 7.8 mg/L at 120 min. It may be attributed to the good mineralization of AB1 molecules into CO₂ and H₂O, while others are converted into more resistant organic compounds that resist degradation by active radicals⁵⁵. Adekunle et al.⁵⁶ detected similar results for the mineralization rate of Eriochrome black T using the Enhanced Nickel Oxide system.

To evaluate the biodegradability of photocatalyzed AB1, the ratios of BOD₅ to COD (BOD₅/COD) and BOD₅ to TOC (BOD₅/TOC) were examined for the AB1 solution, as presented in Fig. 8b. The figure illustrates an improvement in the BOD₅/COD ratio from 0.26 to 0.72 and for BOD₅/TOC from 0.745 to 1.38 with the extension of irradiation time (0 to 120 min). As the threshold values for the BOD₅/COD and BOD₅/TOC ratios are 0.4 and 0.8, respectively, the treated solution meets such criteria⁵⁷. Furthermore, the results highlight that the MCTA process effectively converts AB1 into biodegradable compounds, particularly with reaction times exceeding 60 min. In line with earlier studies, the confirmation of effluent biodegradability can be achieved through the utilization of the carbon oxidation state (COS) and average oxidation state (AOS) techniques, as specified in Eqs. (7) and (8)⁵⁸. In the context of these equations, COD signifies the chemical oxygen demand concentration at reaction time *t* (mg/L), TOC refers to the initial total organic carbon concentration (mg/L), and TOC_{*t*} indicates the concentration of soluble TOC at reaction time *t* (mg/L). Figure 7c displays graphs showing the COS and AOS values of the effluent solution over various reaction times. Over a reaction time of 120 min, the AOS and COS values gradually rise from -0.27 and +1.64 to 1.75 and 3.93, respectively. This trend indicates the transformation of AB1 molecules into degradable compounds through the photocatalytic process, making the effluent suitable for introduction into the biological treatment process.

Halomoan et al.⁵⁷, using CuO-Gd₂Ti₂O₇ for the degradation of malachite green dye, reported removal efficiencies of 92% for TOC and 88% for COD were reported. Moreover, the BOD₅/COD ratio rose from 0.27 to 0.7, signifying the transformation of non-biodegradable wastewater into biodegradable wastewater.

$$\text{COS} = 4 - 1.5 \left[\frac{\text{COD}}{\text{TOC}_t} \right] \quad (7)$$

$$\text{AOS} = 4 - 1.5 \left[\frac{\text{COD}}{\text{TOC}} \right] \quad (8)$$

Catalyst recovery

One of the most important parameters for the economic viability of a catalyst is its recovery and reuse. In this study, recovery and reuse experiments were conducted in five stages. The experiments were performed at the optimum amount obtained from the RSM optimization method. After each use, the catalyst was separated

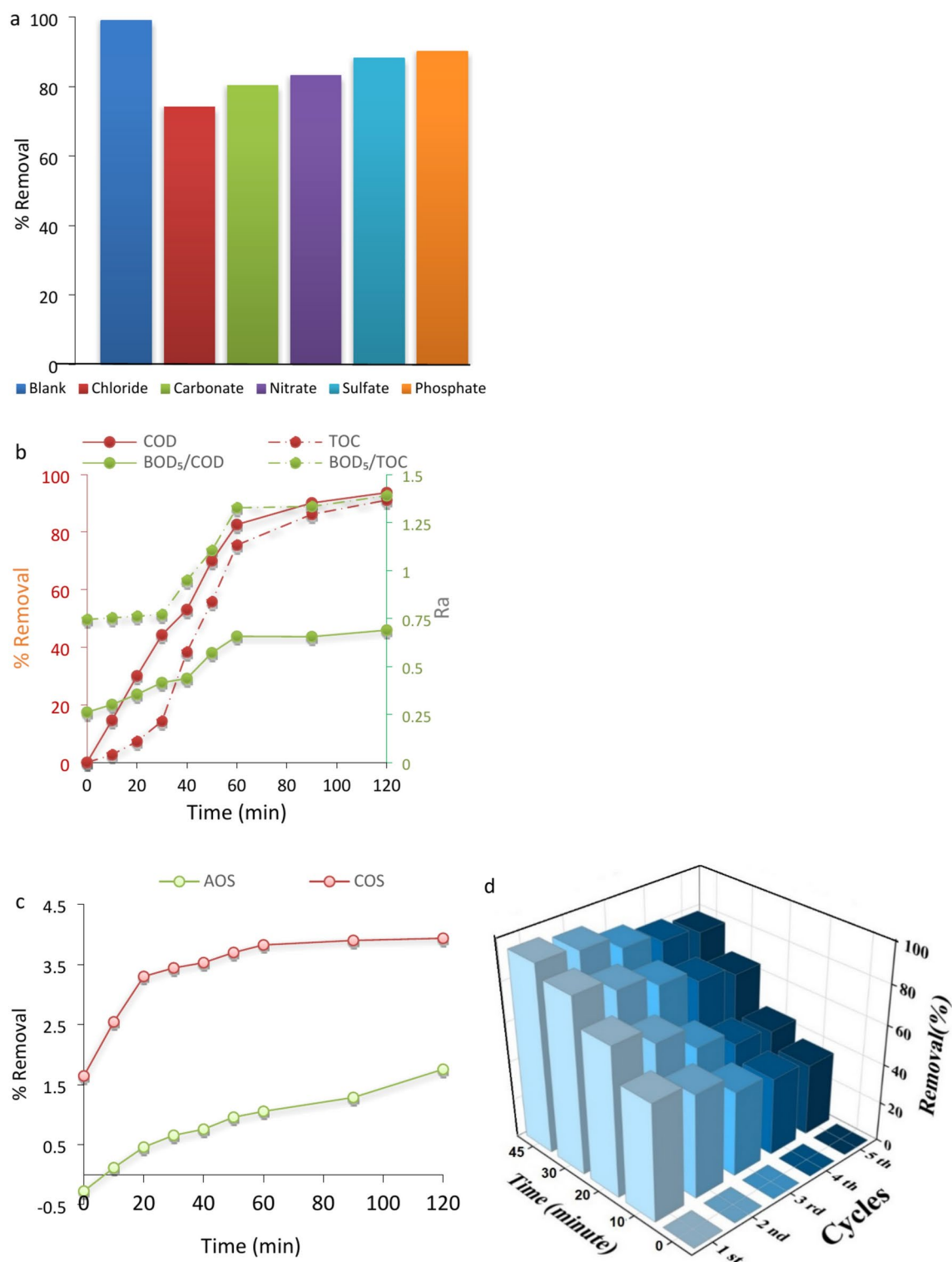


Fig. 8. Effect of anions on the amount of AB1 removal (a); Biodegradability chart (b); AOS and COS ratio in biodegradability process (c); Reusability cycles of MCTA catalyst (d); Antibacterial activities of AB1 after photocatalytic (e) and adsorption process (f).

using iron wool, underwent multiple wash cycles with distilled water, subjected to ultrasonic treatment for 10 min, and finally dried at 105 °C for further testing. After five recovery cycles, the catalyst still maintains its efficiency, although the removal percentage has decreased (Fig. 8d). It can be justified for two reasons. Firstly, the reduction in the catalyst during collection, where approximately a 2% reduction in the collection amount was observed at each stage. Secondly, the saturation of active surface sites of the catalyst during preparation can

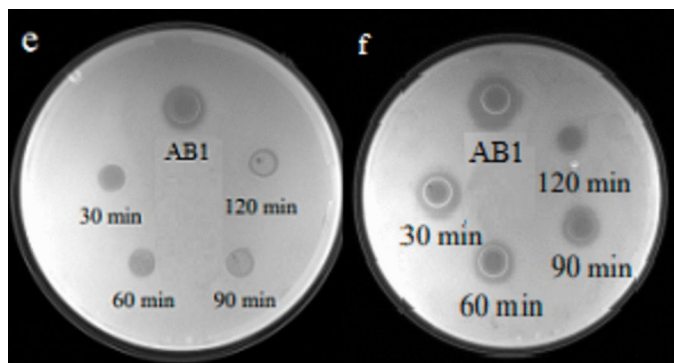


Figure 8. (continued)

be mentioned⁵⁹. This saturation is similar to the study conducted by Hitkari et al.²³, which aimed at Congo red dye removal using copper-zinc oxide nanocomposite, where a reduction was also observed during the recovery stage, with approximately 20% reduction in the weight of the adsorbent during the separation stage, attributed to its magnetic properties in this study.

In the study of Yilmaz et al.²⁰ for removing the AB113 using the photocatalytic process with the $\text{Fe}_3\text{O}_4/\text{ZnO}/\text{GO}$ catalyst, the recovery rate of the catalyst is 95%, and during the five recovery stages, about 25% of the catalyst has been lost, while in this study, 10 percent of the catalyst was reduced during the 5 steps. The decrease in the amount of metal in the period of evaluation can be considered as one of the reasons for the decrease in pollutant degradation.

Toxicity test

To assess the remaining antibacterial effectiveness of the AB1 solution treated with MCTA, a microbiological antibacterial activity assay was conducted using *E. coli* as a reference. In Fig. 8e, it is evident that, with an extended photocatalytic duration, the bacteriostatic ring exhibited a diminishing diameter, reducing from 9.5 mm to 0 mm after 120 min, indicating the loss of antibacterial activity in the residual AB1 within the water environment post-photocatalytic reaction. Figure 8f presents a comparative analysis of the antibacterial test results for AB1 dye treated exclusively with the catalyst through adsorption. Notably, the persistence of the bacteriostatic ring post-adsorption suggests the retention of antibacterial activity in the antibiotics. These findings imply a substantial reduction in the antibacterial properties of AB1 after the photocatalytic reaction, thereby offering notable advantages for the efficient recycling of dye wastewater.

Identification of active species and elucidation of the mechanism

The photocatalytic degradation of AB1 can be ascribed to the joint action of active species, including h^+ , e^- , $\text{O}_2^{\cdot-}$, and $\text{OH}^{\cdot}$. To assess how each active species contributes, scavenging experiments were conducted using isopropanol (IPA) to trap $\text{OH}^{\cdot}$, ammonium oxalate (AO) to scavenge H^+ ions, $\text{K}_2\text{S}_2\text{O}_8$ to capture e^- , and 1,4-benzoquinone (BQ) to quench $\text{O}_2^{\cdot-}$ ⁶⁰. Figure 9a summarizes the results of AB1 degradation experiments with MCTA catalyst in the absence and presence of these scavengers. The introduction of IPA (1 mM) significantly reduces the initial efficiency of 99.1% to 61.2% after 1 h of irradiation. This observation highlights the crucial involvement of $\text{OH}^{\cdot}$ in the degradation mechanism. Similarly, upon introducing 1 mM BQ, the efficiency dropped to 70.3% within 60 min, suggesting that $\text{O}_2^{\cdot-}$ is a key factor in the degradation process of AB1 while the inhibition effects of AO and $\text{K}_2\text{S}_2\text{O}_8$ were relatively minor, suggesting that the involvement of H^+ and e^- in the photocatalytic degradation of AB1 is minimal.

The existence of $\text{OH}^{\cdot}$ and $\text{O}_2^{\cdot-}$ was reaffirmed through the Electron Spin Resonance (ESR) spin-trap technique employing 5,5-dimethylpyridine-N-oxide (DMPO). The achieved ESR spectra using the MCTA/sunlight system (Fig. 9b) exhibited characteristic signals for DMPO- $\text{OH}^{\cdot}$ and DMPO- $\text{O}_2^{\cdot-}$ spin adducts, further substantiating the involvement of these radicals in AB1 degradation⁶¹. Therefore, the proposed photocatalytic system primarily relies on $\text{O}_2^{\cdot-}$ and $\text{OH}^{\cdot}$ for AB1 degradation, with supplemental contributions from photogenerated holes.

Based on DRS, trapping, and EPR spectra experiments, Fig. 8c displays a proposed schematic diagram elucidating the potential mechanism for the photocatalytic degradation of AB1. Before interpreting the possible mechanism, the conduction band (CB) and valence band (VB) energies of TiO_2 and Fe_3O_4 particles were determined through Eqs. (9) and (10)⁶¹.

$$E_{\text{CB}} = \chi - E_{\text{C}} - 0.5E_{\text{g}} \quad (9)$$

$$E_{\text{VB}} = E_{\text{CB}} + E_{\text{g}} \quad (10)$$

The CB potential of the semiconductor is denoted by E_{CB} , while E_{VB} represents the VB potential of the semiconductor. The electronegativity of the Mulliken catalyst is represented by χ , with values of 5.78 and 5.81 eV for Fe_3O_4 and TiO_2 , respectively^{62,63}. E_{C} symbolizes the energy of free electrons at 4.5 eV on the hydrogen scale, and E_{g} represents the energy associated with the band gap of particles.

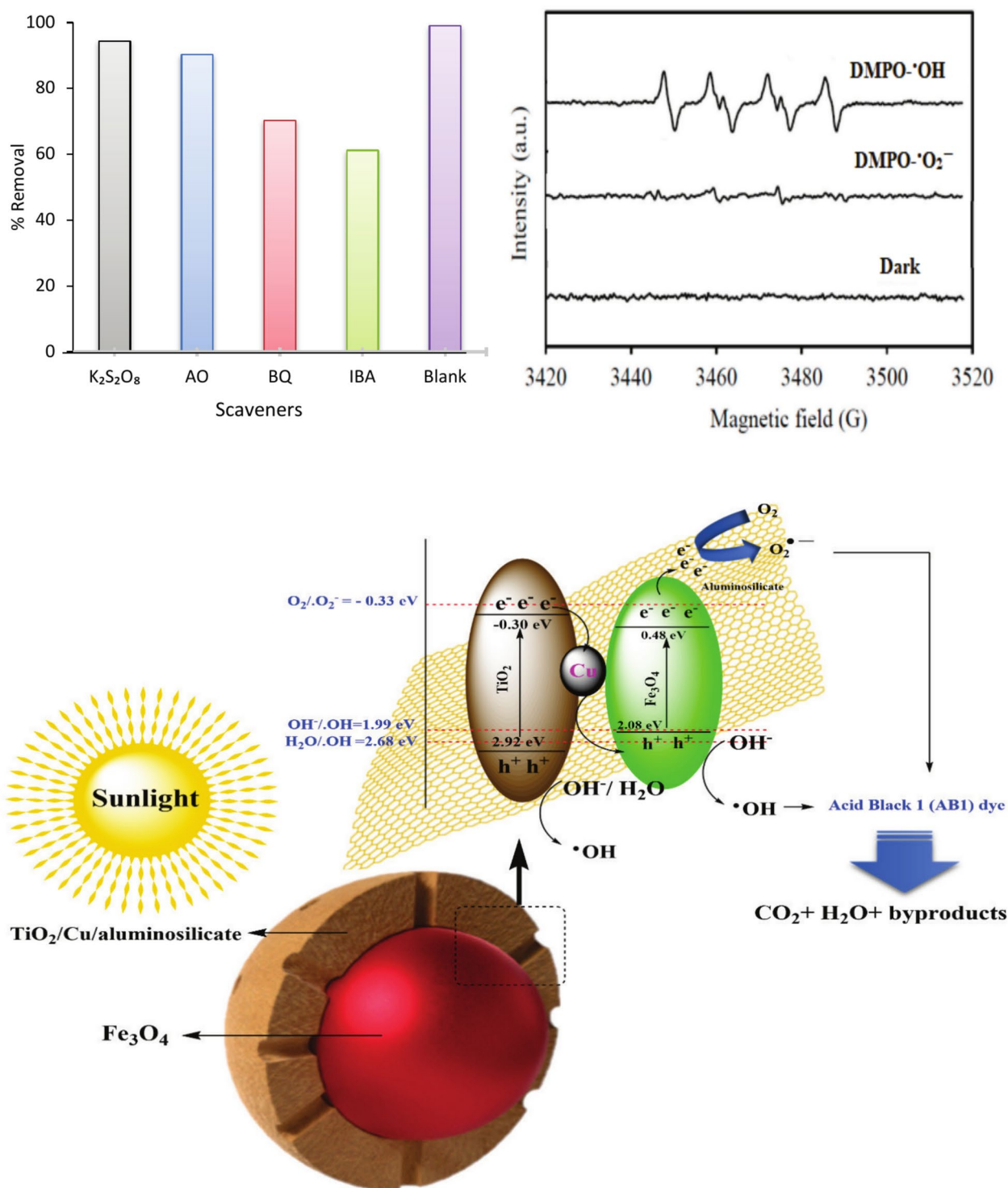


Fig. 9. Effect of quenching agent on degradation (a); ESR spectra of DMPO-•O₂⁻ and DMPO-•OH (b); Schematic representation of the pollutant degradation mechanism by the photocatalysis process based on MCTA (c).

According to Eg TiO₂ (3.22 eV) and χ stated in the earlier study⁶³ (5.81 eV), E_{CB} and E_{VB} were calculated to be -0.3 eV and 2.92 eV, respectively. For Fe₃O₄, E_{CB} and E_{VB} values reported by Channei et al.⁶¹ were used (+0.48 eV and +2.08 eV, respectively). When the MCTA nanocomposite is exposed to sunlight, every component of the MCTA nanocomposite becomes excited, giving rise to the production of e⁻ and h⁺. The photogenerated e⁻ is transferred from the TiO₂ conduction band to the Cu surface because the Cu surface is more positive than the TiO₂ nanoparticle CB. Next, some of e⁻ is directed from Cu to the Fe₃O₄ valence band, owing to the lower positioning of the Fermi level of Cu metal relative to the energy of the valence band⁶². This event is beneficial

Process	Conditions	% Removal	Refs.
Chitosan/Fe ₂ O ₃	pH = 5, AB113 = 25 mg/L, Time = 90 min	84.1%	2
]Fe ₃ O ₄ /ZnO/GO	pH = 3, dose = 0.8 g/L, Time = 75 min	89.2%	20
TiO ₂	pH = 5, MB = 50 mg/L, Time = 60 min	81.2%	21
Zinc-sulfide/RGO	pH = 3, AB1 = 10 mg/L, Time = 45 min	92.3%	22
CeO ₂ /Ce ₂ S ₃	pH = 3, dose = 1 g/L, Time = 45 min	93.6%	24
TiO ₂ /GO	pH = 5, MB = 25 mg/L, Time = 75 min	92.2%	29
TiO ₂ -x/g-C ₃ N ₄ /CNFe	pH = 5, C ₀ = 25 mg/L, Time = 60 min	94.3%	42
MMT-CuFe ₂ O ₄	pH = 5, C ₀ = 50 mg/L, Time = 45 min	94.2%	50
MCTA	pH = 3, AB1 = 37.5 mg/L, Time = 45 min, dose = 0.75 g/L	99.1%	This study

Table 3. A detailed comparison of outcomes derived from oxidation experiments on the solution.

for the photocatalysis process through declining e^-/h^+ recombination. In this way, the photogenerated electron in Fe₃O₄ is transferred to the aluminosilicate surface. The electron on the aluminosilicate surface converts the O₂ molecule into the O₂^{•−} radical⁶³. The holes produced in VB TiO₂ and VB Fe₃O₄ can oxidize H₂O or OH[−] molecules to generate [•]HO because the positive value of VB particles is greater than the potential of H₂O/HO[•] (E₀ = +2.34 eV vs. NHE) and OH[−]/HO[•] (E₀ = +1.99 eV vs. NHE)⁶⁴. Finally, all the active species produced can decompose AB1 in two phases, soluble and solid, into CO₂, H₂O compounds and low biological risk products.

Comparison with studies

To evaluate the effectiveness of the synthesized catalyst, its performance was compared with results from prior studies (Table 3). The analysis revealed that all the examined processes could achieve dye removal efficiencies exceeding 80%. However, this high efficiency was obtained under a range of laboratory conditions, complicating direct comparisons among the findings. Notably, textile industry wastewater generally exhibits a low pH, and conventional wastewater treatment plants have shown limited success in dye removal. Table 3 indicates that acidic environments and increased catalyst dosages enhance the efficiency of most advanced oxidation processes. The MCTA photocatalytic system demonstrated complete AB1 removal under acidic conditions (pH 3). Achieving 100% AB1 degradation within a mere 60 min, our catalyst surpassed the efficiency of other semiconductor-based photocatalytic systems. These results highlight the superior ability of our photocatalyst to produce reactive species necessary for the efficient breakdown of AB1.

Conclusion

The analysis of the prepared photocatalytic nanocomposite involved a comprehensive examination using various microscopic and spectroscopic methods. By assessing the optimal nanocomposite sample, it was tested as a nanostructured catalyst for the photodecomposition of AB1. Trapping experiments demonstrated that hydroxyl radicals exhibited superior efficacy in decomposing AB1 molecules when facilitated by the MCTA nanophotocatalyst. The Box–Behnken design disclosed that all the main factors were statistically significant, with the catalytic dosage having the most significant impact (F = 854.2), and the initial pH exhibiting the least impact (F = 17.2) among the main factors. Furthermore, in this study, catalyst recovery experiments were conducted, and the findings demonstrated a consistently high efficiency of the catalyst over five consecutive cycles. Also, the results of the biodegradability tests showed that the BOD₅/COD ratio improved from 0.26 to 0.72, and for BOD₅/TOC, it was 0.745–1.38 as the duration of irradiation extended from 0 to 120 min. This indicates the process of transforming wastewater from a non-biodegradable to a biodegradable composition. Furthermore, as the photocatalytic duration extended, the bacteriostatic ring diameter progressively diminished from 9.5 to 0 mm within 120 min. Analysis of the impact of anions indicated a subtle inhibition in AB1 degradation, resulting in extended reaction times. The findings from this experimental study underscore that incorporating Cu/aluminosilicate into TiO₂, coupled with the hydrothermal and solvothermal synthesis of the MCTA nanocomposite, yielded an innovative high-performance nanocatalyst for applications in water remediation and environmental purification.

Data availability

The datasets used and/or analyzed during the current study are available from the corresponding author on reasonable request.

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Author contributions

FM: analysis of colored wastewater, NA and NM: Writing—original draft, Visualization. DB: Writing—review & editing, Visualization. MA: analysis of catalyst characteristics.

Declarations

Competing interests

The authors declare no competing interests.

Ethics approval

No artifacts or human participants are involved in this study.

Additional information

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Correspondence and requests for materials should be addressed to D.B.

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